

Chittagong University of Engineering and Technology
Department of Computer Science and Engineering

Development of a Machine Learning Model for Barbell Exercise Classification and Repetition Counting Using Accelerometer and Gyroscope Data

BY
Md. Razuanul Haque
ID: 1904104
Abdullah Al Mubin
ID: 1904105
Mohammad Sydul Hasan
ID: 1904127
Faisal Ahammad Shawon
ID: 1904128

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Chittagong University of Engineering and Technology (CUET)
Department of Computer Science and Engineering
Chattogram-4349, Bangladesh.

Final Report of S/W Development Project

Student Name and ID
Md. Razuanul Haque
190104
Abdullah Al Mubin
1904105
Mohammad Sydul Hasan
1904127
Faisal Ahammad Shawon
ID: 1904128

Course Teacher

Animesh Chandra Roy
Assistant Professor, Department of CSE, CUET

MD. Atiqul Islam Rizvi
Assistant Professor, Department of CSE, CUET

Project Title

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1 Abstract

Strength training, in addition to aerobic exercises, is an important component of a balanced exercise program. However, mechanisms for automatically tracking free weight exercises have not yet been fully studied. This paper aims to explore further the possibilities of context-aware applications within the strength training domain. It does so by analyzing wristband accelerometer and gyroscope data obtained during strength training sessions. The collected dataset contains data from 5 participants performing various barbell exercises. The goal is to explore, build, and evaluate models that can, just like human personal trainers, track exercises, count repetitions, and detect improper form. The methods evaluated in this paper use a supervised learning approach for classification. Various machine learning algorithms were trained using the collected dataset and accuracies were compared to find and evaluate the right models.

2 Introduction

During the last decade, many practical constraints related to carry-on sensors like accelerometers, gyroscopes, and GPS receivers have been solved. This has enabled monitoring and classification of human activity based on information from wearables such as smartwatches to become a growing research area of pattern recognition and machine learning. The grounds for that lie in the high commercial potential of context-aware applications and user interfaces. Moreover, activity recognition can be utilized to attack some of the serious societal challenges like rehabilitation, sustainability, elderly care, and health.

To promote healthier lifestyles works from the past have focused on tracking movements and user feedback via exercise management systems. Such systems partly replace tasks that are currently fulfilled by personal trainers. For example, in the category of aerobic exercises such as bicycling, swimming, and running, there are accelerometers and GPS-based pedometers to track running pace and distance, ECG monitors to track exertion, and electronic exercise machines such as treadmills, elliptical trainers, stair climbers, and stationary bikes. Weight training, in addition to aerobic exercises, is an important component of a balanced exercise program [2]. However, mechanisms for tracking free weight exercises have not yet been fully studied. Currently, there is only one fitness wearable brand on the market that claims to automatically identify exercises and record repetitions [2]. However, the available literature on context-aware applications within the strength training domain is still limited which is probably due to the commercial potential of such methods.

With advancements in context-aware applications, it should eventually be possible to create fully digital personal trainers. To establish the requirements of such a digital trainer, the roles of current personal trainers will be discussed briefly. First of all personal trainers must have knowledge of human anatomy and fundamental principles of exercise science and nutrition. Secondly, the ability to design and execute exercise programs, tailored to the needs and attainable goals of the individual in an effective and motivational way. Finally, a way of tracking workouts to ensure appropriate form and progressive overload is applied to exercises to help clients safely reach their fitness goals [3].

While recent years have seen great improvements in digitalizing the first two roles of the personal trainer, the final, and most important role in terms of safety and progress, is not well implemented in today's fitness wearables and applications yet. Therefore,

this paper aims to explore further the possibilities of context-aware applications within the strength training domain. The methods evaluated in this paper are based on work from Hoogendoorn and Funkuse [2] and use supervised learning approaches for classification. Various machine learning algorithms were trained using the collected dataset and accuracies were compared to find the right models.

3 Experimental Setup

The program is made up of 5 fundamental barbell exercises: Bench Press, Deadlift, Overhead Press, Row, and Squat. The exercises are shown in Figure 1.

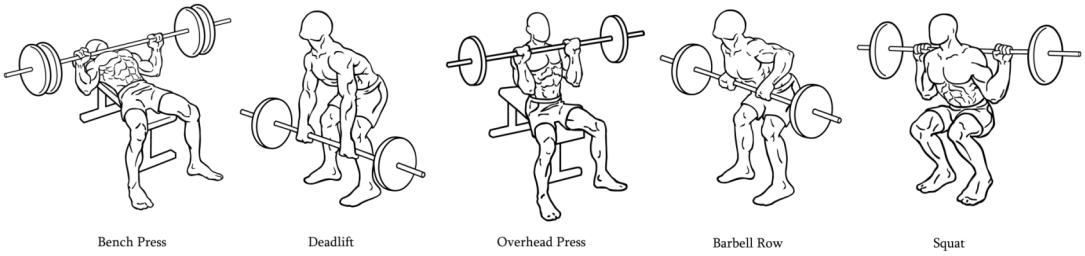


Figure 1: Basic Barbell Exercises

In strength training, a program is described using the terms sets and repetitions (reps). A rep is one complete motion of an exercise and a set is a group of consecutive repetitions followed by a resting period. In the Starting Strength program, exercises are performed with a heavy weight. The aim is to pick a weight(Heavy and lightweight based on participants) so that around 5 repetitions can be done.

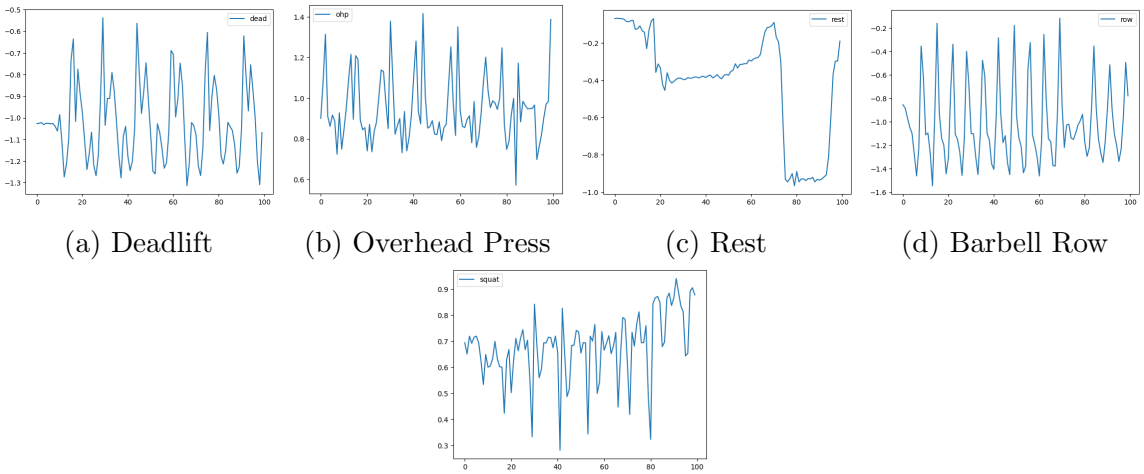
3.1 Data Collection

A wristband sensor was used to collect data for this experiment. The wristband mimics the placement and orientation of a watch while allowing for controlled experiments. Data was collected using the default settings of the sensors: accelerometer: 12.500HZ and gyroscope: 25.000Hz. To collect a representative dataset, 5 participants were asked to perform the barbell exercises in 3 sets of 5 repetitions. In another training session, the participants were asked to perform the same exercises but in 3 sets 10. These sets of higher reps will be used to see how models generalize over different weights. This resulted in a total of 150 (5x5x6) sets of data [1]. Additionally, in between some sets 'resting' data was collected. To keep the data as representative as possible, the test subjects had no restrictions during the resting periods. Therefore, the resting periods are a mix of standing, walking, and sitting. This data will be used to examine the state change from rest to exercise.

4 Converting Raw Data

The raw dataset contained a total of 69677 entries, each containing an epoch timestamp and x, y, and z-values from the sensor. The individual sensor measurements from

the wristband are split up into different files where each entry has a unique timestamp. Because of this, the data had to be aggregated. To minimize information loss in this process, a small step size of ∇ t = 0.20 s, i.e. five instances per second was chosen. Numerical values were aggregated using the mean, and categorical attributes (labels) were aggregated using the mode. The aggregated dataset provided us with a good starting point for visualizations. Figure 2 shows accelerometer data for a heavy set of each exercise. The figure shows that exercise patterns are pretty unique and the repetitions can easily be identified from the peaks. Figure 3 shows the differences in y-acceleration between medium and heavy squat sets. The medium-weight sets have higher peaks whereas the heavy sets have deeper drops. This makes sense as a medium-weight reps go up faster due to lower resistance while the heavy reps drop down faster due to the heavy load.



(a) Deadlift (b) Overhead Press (c) Rest (d) Barbell Row

(e) Squat

Figure 2: Accelerometer Data

This section will continue to explain approaches for handling raw, noisy data. The goal is to transform the data in a way that subtle noise is filtered and the parts of our data that explain most of the variance are identified. Two approaches will be discussed: Low-pass Filtering, which can be applied to individual attributes, and Principal Component Analysis, which works across the entire dataset.

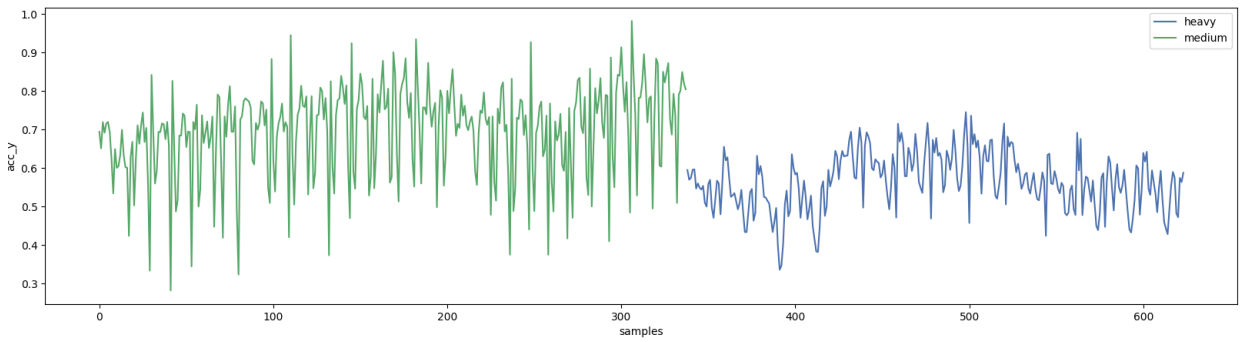


Figure 3: Medium and heavy weight squats

4.1 Low-pass Filter

The low-pass filter can be applied to data that is of temporal nature, and assumes that there is a form of periodicity. The Butterworth low-pass filter can remove some of the high-frequency noise in the dataset that might disturb the learning process [3]. The filter was applied to all but the target features. The visualizations showed that the movements have a frequency of around 2 seconds per repetition. After some further visual inspection of some trial-and-error results as suggested in the work of van den Bogert [1], the cut-off point was set at 1.2 Hz. Figure 4 shows the y-acceleration for a heavy bench press set before and after smoothing

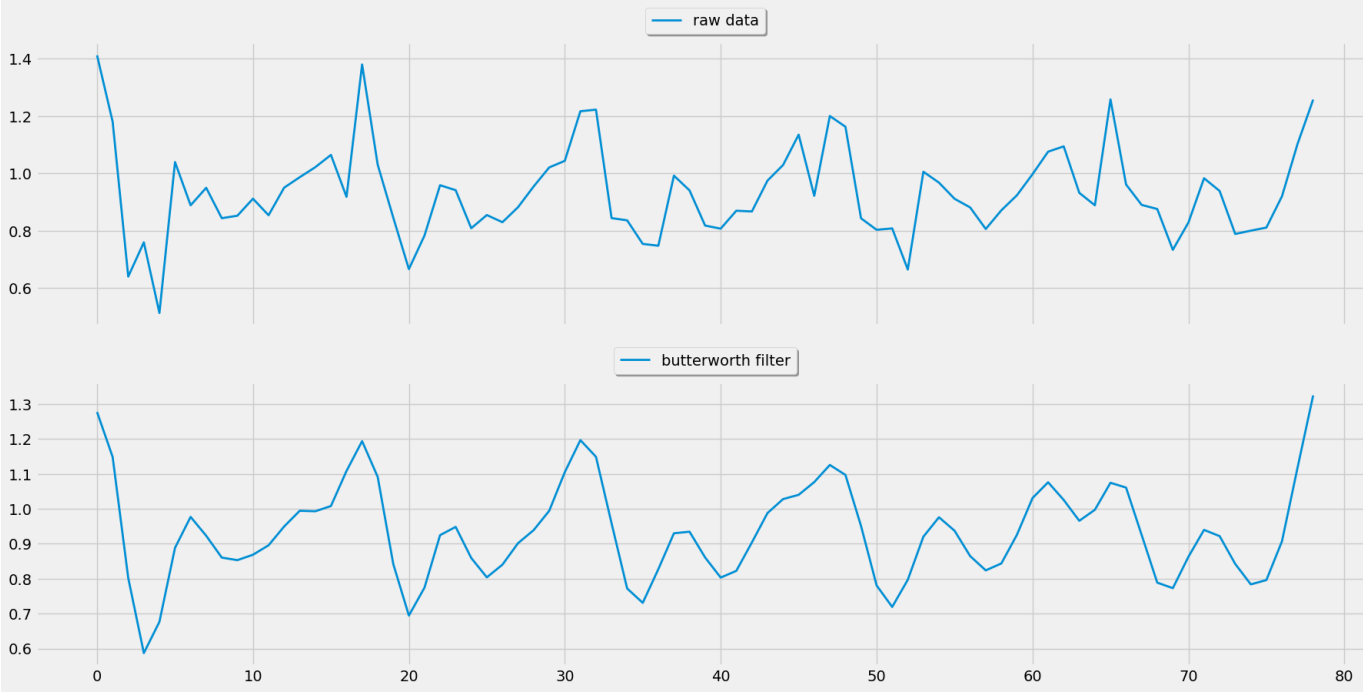


Figure 4: Low-pass Filter

4.2 Principal Component Analysis

A principal component analysis (PCA) was conducted to find the features that could explain most of the variance. PCA was applied to all features excluding the target columns. The results are visualized in Figure 5 which shows that the explained variance drastically decreases after 3 components. Therefore, 3 components are selected and their values are included into the dataset.

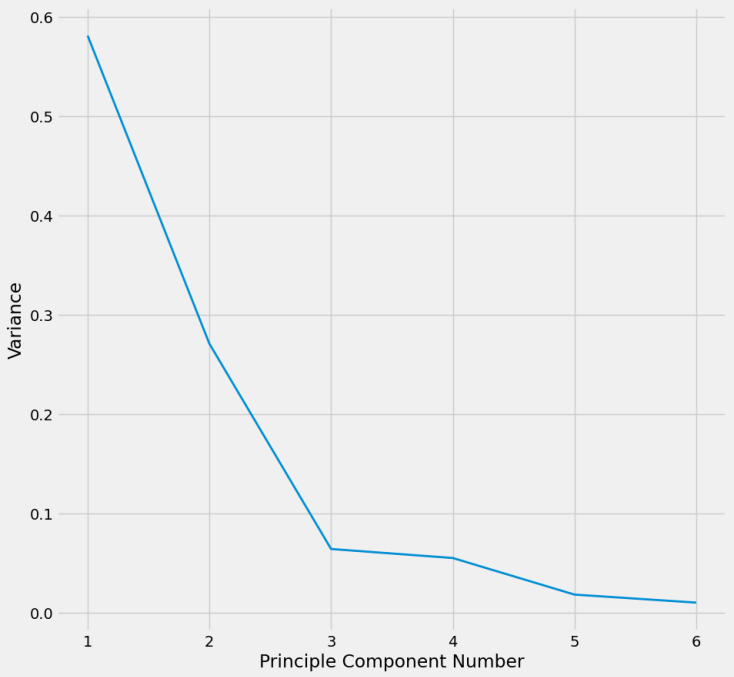


Figure 5: Principal Component Number

5 Feature Engineering

In this section, we will discuss how additional features were derived from the original dataset including aggregated features, time features, frequency features, and clusters.

5.1 Aggregated Feauters

To further exploit the data, the scalar magnitudes r of the accelerometer and gyroscope were calculated. r is the scalar magnitude of the three combined data points: x, y, and z. The advantage of using r versus any particular data direction is that it is impartial to device orientation and can handle dynamic re-orientations. r is calculated by:

$$\mathrm{r}_{\mathrm{magnitude}} = \sqrt{x^2 + y^2 + z^2} \tag{1}$$

5.2 Time Domain

To exploit the temporal nature of the data, numerical data points were aggregated by taking the standard deviation (sd) and the mean of all features except the target columns over various window sizes. The sd should capture the variation in the data over time. For example, the sd is expected to be higher during exercises than during resting periods. The temporal mean will indicate more about the general levels of the data, with less influence from spiky noise. When selecting an appropriate window size, one should balance how strongly the data would be affected by noise on the one hand, and maintaining predictive power on the other hand. Figure 6 shows the results for window size 5. A window size of 5 was chosen and added to the dataset.

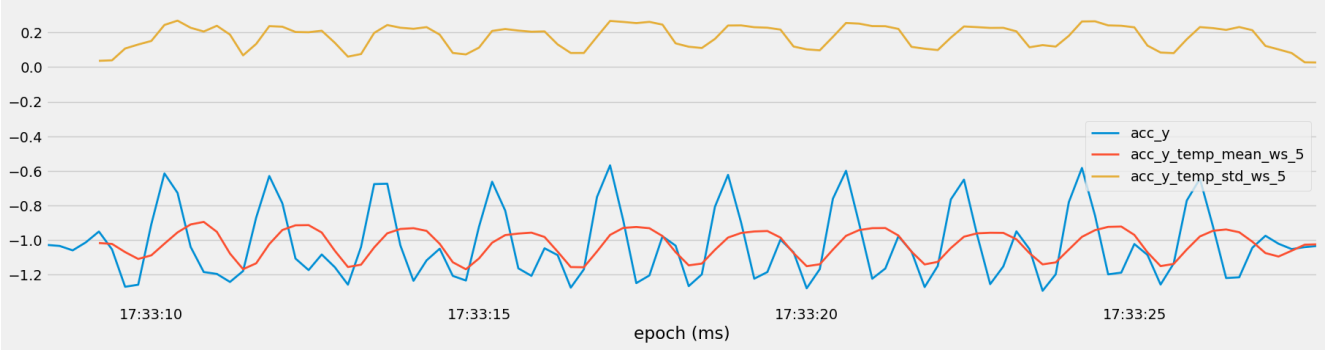


Figure 6: Numerical temporal aggregation with a window size of 5 for acc_y

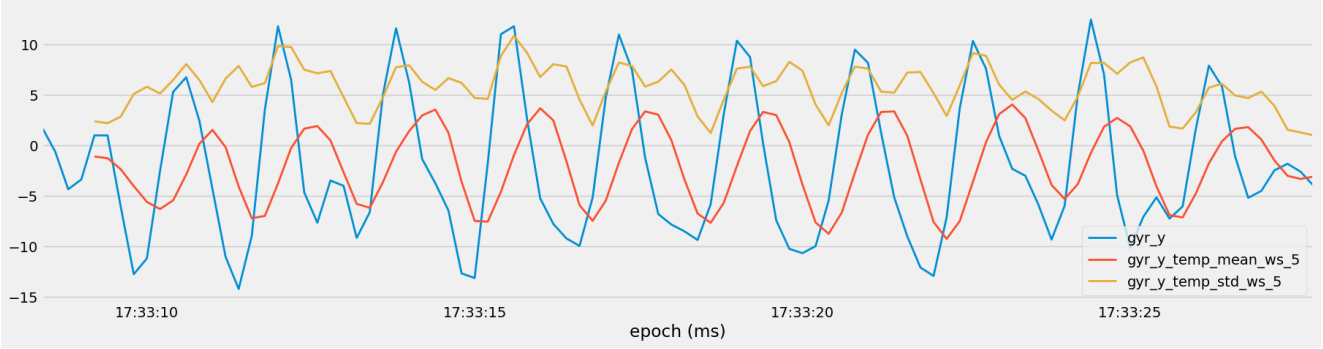


Figure 7: Numerical temporal aggregation with a window size of 5 for gyr_y

5.3 Frequency Domain: Fourier Transformation

Besides the time domain, the frequency domain will also be explored. The idea of a Fourier transformation is that any sequence of measurements can be represented by a combination of sinusoid functions with different frequencies [1]. The same window size of 14 was used to compute the frequency features with a Fourier Transformation including the maximum frequency, the frequency signal weighted average, and the power spectral entropy [1].

5.4 New Dataset

The dataset now contains various new features. However, since there are overlapping time windows, the resulting attributes are highly correlated. Given this overlap, just including all instances might not provide us with new information as only one point in the window differs for adjacent instances. To prevent overfitting, a maximum overlap for the windows was set accordingly, and instances for which this criterion was not met were removed. 45% overlap was allowed for. This reduced the dataset to 3868 instances. Some information is lost in this process but it has been shown to pay off as there are less very similar instances in the dataset that could cause overfitting [2].

5.5 Clustering

It is possible that a membership to a certain cluster can help predict a label. After running some tests, k-means clustering (k=5) was preferred by elbow technique. The

results are shown in Figure 8. Cluster 2(orange) covers almost all the bench press and overhead press data. This makes sense as both pressing movements are very similar. The squat is captured in cluster 3(yellow) and the deadlift and row are captured in cluster 4(olive) with a perfect score. Cluster 6(violet) contains some of the rest data but fails to capture this accurately.

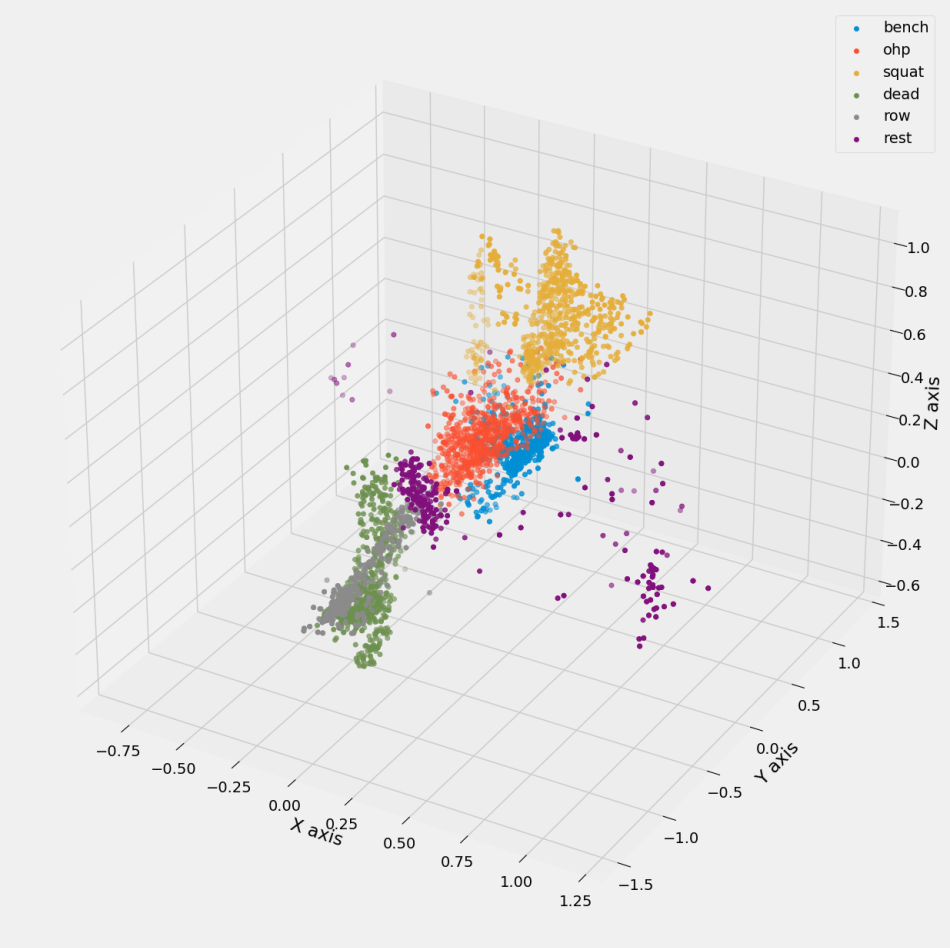


Figure 8: Clusters

6 Modeling

The dataset is now processed and ready for training. The dataset contains the 6 basic features, 2 scalar magnitude features, 3 PCA features, 16 time features, 12 frequency features and 1 cluster feature. This section will explain how the models for classification, and repetition counting is evaluated.

6.1 Classification

Because of the temporal nature of our dataset, the training and test set were split up based on the exercise sets. The training data contains 75% of the whole data set, and the test data was 25% of the whole data set.

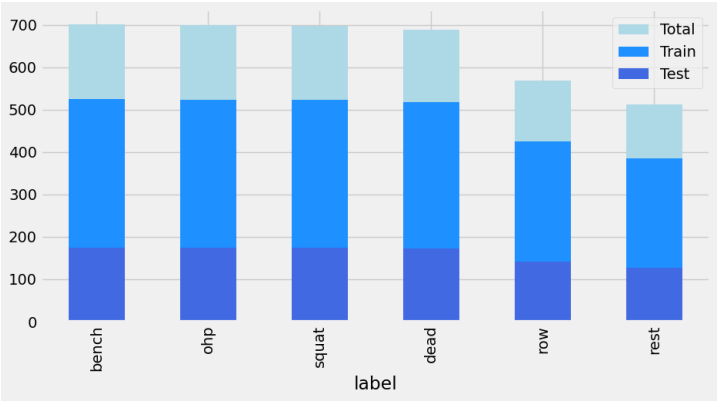


Figure 9: Splitting of the Data Set

Feature Selection: Forward feature selection was used to investigate which features contribute the most to performance as useless features could impact the performance of the algorithms. Using a simple decision tree, and gradually adding the best features, the results showed us that after 10 features the performance no longer significantly improved. And the best 10 features are: *acc_z_freq_0.0_Hz_ws_14*, *acc_x_freq_0.0_Hz_ws_14*, *duration*, *acc_y_temp_mean_ws_5*, *gyr_r_freq_0.0_Hz_ws_14*, *gyr_x_freq_178.571_Hz_ws_14*, *gyr_y_temp_mean_ws_5*, *gyr_z_freq_107.143_Hz_ws_14*, *acc_y_freq_250.0_Hz_ws_14*, *acc_y_freq_35.714_Hz_ws_14*.

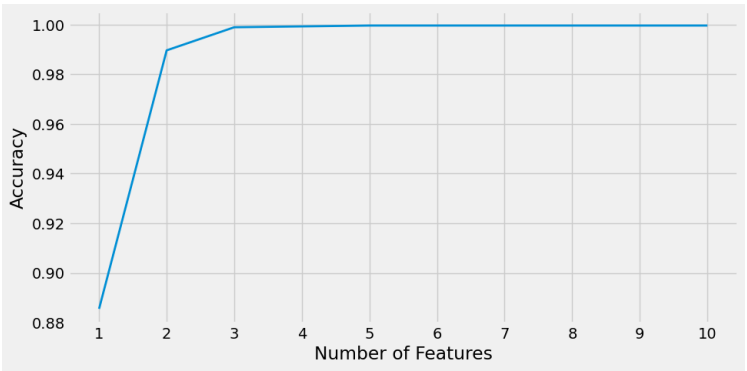
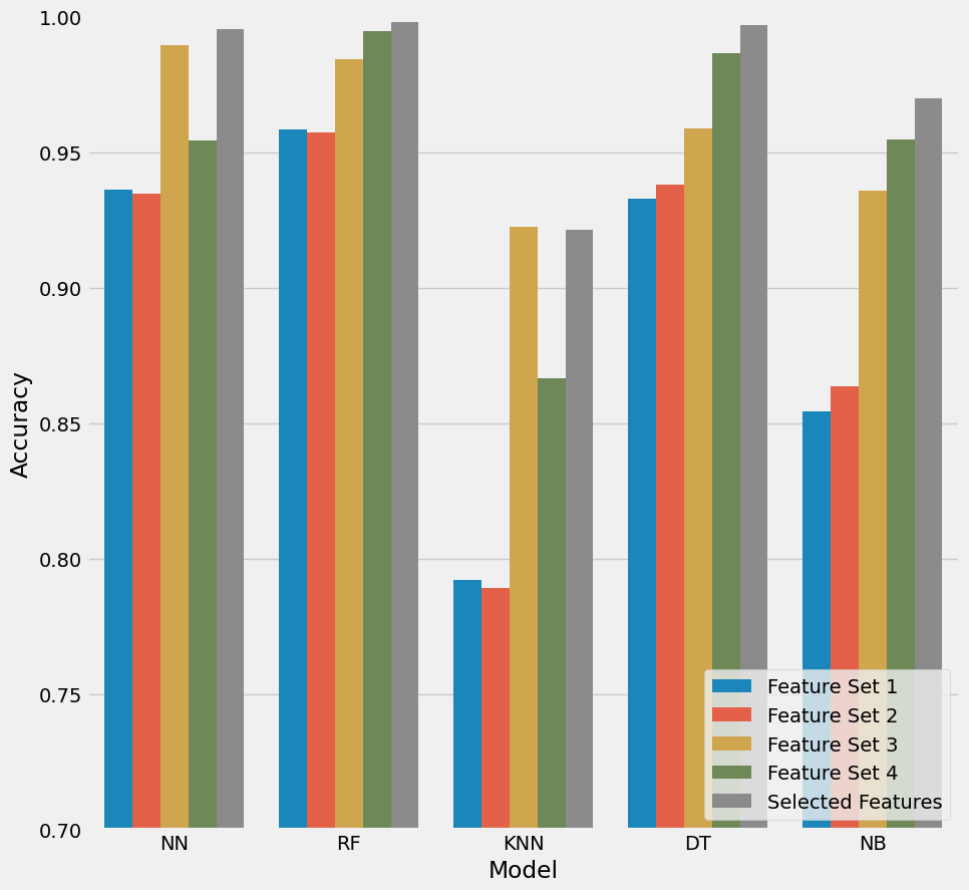


Figure 10: Accuracy of Features

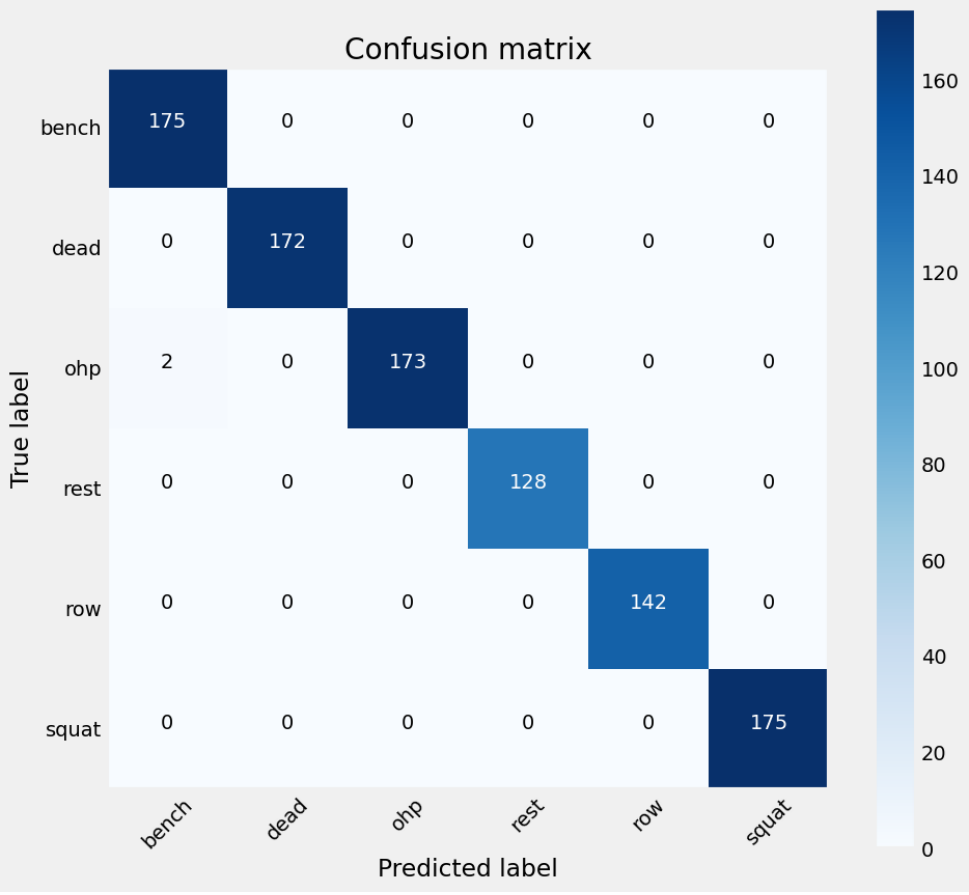
Models: First, an initial test run was done to determine the performance of a selection of models and features. This test included the following models: Neural network, Random Forest, Support Vector Machine, K-nearest Neighbours, Decision Tree, Naive Bayes. A grid search was performed on all of the models.

6.2 Classification Results

We ran several models with all possible features. From them, Random Forest has given the maximum accuracy, which ran under the 10 best features we've got from the feature selection that can be seen in Figure 11(a). Using a 5-fold cross-validation upon which grid search parameter tuning was performed, the following optimal parameters for the Random Forest was found: minimum samples per leaf: 2, n-estimators: 100, criterion: Gini. A confusion matrix of the final predictions is shown on the right in Figure 11(b). The model achieved an overall accuracy of 0.997932% with similar precision, recall, and f1-score on the test set.



(a) Accuracy of different models with different features



(b) Confusion Matrix

Figure 11: Model Performance and RF Classification Confusion Matrix

6.3 Counting Repetitions

To count repetitions, a simple peak counting algorithm was applied to the scalar magnitude acceleration data. To make sure small local peaks were neglected, a strong low-pass filter with a cut-off at 0.4 Hz was applied first. It was found that this method of counting repetitions has to be adjusted to the individual exercises for the best performance. For the deadlift and overhead press, better results were obtained by counting minimum values. The overall error rate for counting repetitions was about 1.02% for the collected dataset. An example of 10 deadlift repetitions with heavy weight is shown in Figure 12.

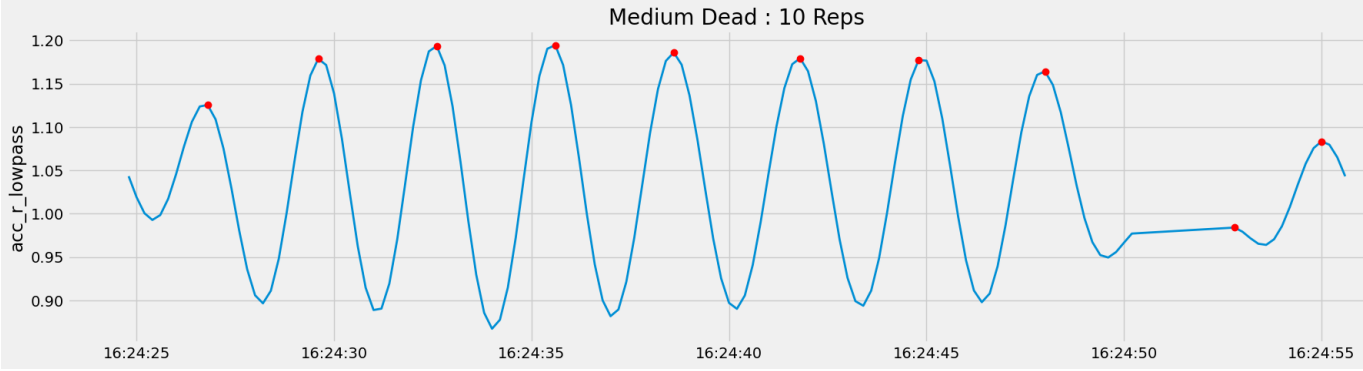


Figure 12: Counting deadlift repetitions after applying a lowpass filter

In Figure 13 we can also see the difference between actual reps versus predicted reps for each exercise with medium and heavy weight.

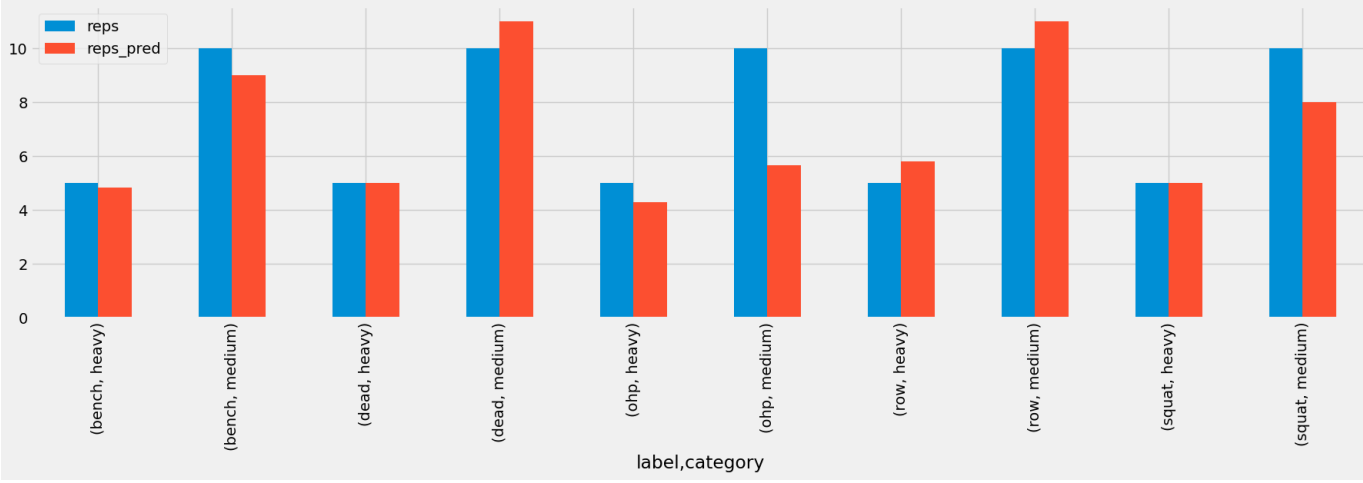


Figure 13: Repetition Counting

6.4 Generalization

It is important to evaluate how the models generalize to new data. Since we have added a lot of time variables in our dataset there is still some overlapping of data. So it could be possible that our model can train and test data over the same exercise. To validate these we again split the dataset based on participants. Now we trained our previously selected model (RF with Selected Features) with the data where the participant

is not "A" and tested with the data where the participant is "A". In this, we got slightly less accuracy from our previous split which is about 99.01%.

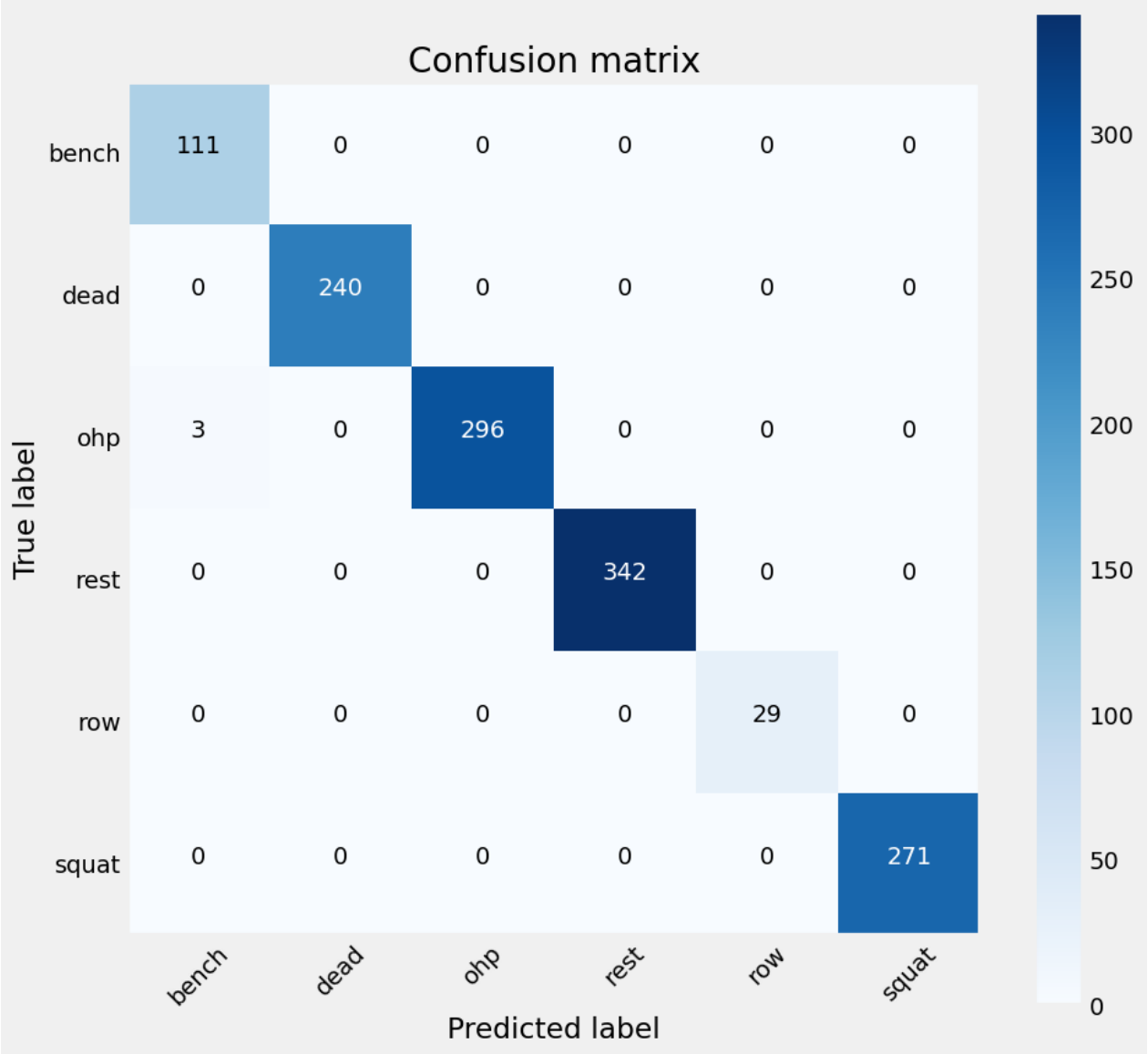


Figure 14: Confusion Matrix after split of data based on participants

7 Conclusion

This project aimed to explore the possibilities of context-aware applications within the strength training domain.

Though our model has a success rate greater than 99% but the model could not achieve a perfect score because it wrongly classified some bench press instances as overhead press and the other way around. A possible explanation for this is that both the bench press and overhead press are vertical pressing movements. While the torso is in a different position, the wrist orientation is similar. A simple peak counting algorithm, in combination with a strong low-pass filter, was applied to the data to count repetitions. It was found that, in order to achieve the best results, rep counting models should be tweaked per exercise. While the method is very basic, it achieved decent results with an

error rate of 1.02%.

Finally, the project can be useful for developing a variety of new applications, such as fitness-tracking apps, exercise learning systems, and injury prevention systems.

References

[1] *Dataset Source, Available Link:Dave Ebbelar GYM Dataset.*

[2] Mark Hoogendoorn and Burkhardt Funk. "Machine learning for the quantified self". In: *On the art of learning from sensory data* (2018).

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